

Regimes' Characteristics and Time Series Forecasting

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Dei uma grande reorganizada no texto seguindo as mudanças que falamos. Na minha visão tem duas coisas separadas: uma é framework do trabalho, seja a parte teórica de definir o DGP geral e o conceito de regime-conditional metrics, seja o framework das simulações; outra é a parte aplicada, seja os DGPs, modelos, e métricas específicas que eu vou considerar, seja os parâmetros e diagnósticos das simulações realizadas. Então organizei dessa forma, mas também daria para organizar como “Theoretica framework” -> “Considered DGPs, ...” -> “Simulation framework” -> “Simulation implementation”.

Mais comentários ...

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1 Introduction

Regime switching (RS) models describe time series that exhibit different behavior – different parameters – across different regimes. They are useful to capture non-linearities in time series, having been widely used in economics and finance, for instance, to model business cycles and market volatility. There are several types of regime switching modeling, some with stochastic switching, such as Markov-switching models, and some with deterministic switching, such as threshold models.

As with any forecasting model, it is important to understand the factors that influence their performance, and how econometricians can use this knowledge to improve their models. In this work, I focus on: (i) the ability of these models to learn and generate performant forecasts under the presence of mis-specification; and (ii) how that ability relates to the characteristics of the regimes by them.

The first focus is common in forecasting econometrics: exactly identifying the data generating process (DGP) is the exception, not the rule, so the modeling goal is actually to find a robust approximator. It is important to document how each RS model behaves under different mis-specifications, explore possible universal approximators, and understand how each element of the DGP affects the learning problem of the models.

The second focus is less orthodox and specific to RS models. These models are special in the sense that they not only identify the series in question but also its states – its regimes – thus allowing the econometrician to describe the distribution of each regime and how different they are from each other. This characterization of regimes’ distributions might be informative for the model’s performance: for example, if the DGP implies different intercepts across regimes, a model whose identified regimes have the same conditional average is probably not capturing that dynamic well; or some class of model can be good at capturing that dynamic but bad at capturing changes on the persistence. These examples might seem obvious, but I will show that there is many useful information to be taken from this kind of analysis.

The nature of this project is explorative. I will simulate a diverse set of DGPs and try to find stylized facts about how each RS model adjusts to them, and how the characteristics of the estimated regimes relate to this adjustment. To make things more concrete, in the remainder of this section I synthesize the methodology, describe the patterns I hope to find, and present some of the actual findings. Additionally, I briefly present the literature on RS models and how my work contributes to it.

1.1 Basic methodology and hypothesis

The methodology follows a common setup. The first step is to establish a theoretical framework that describes all RS models in a unified way. Here, I denote the separate ‘ingredients’ in an RS DGP: the *series generating process* (SGP) and the *regime generating process* (RGP). By varying these ‘ingredients’, I define a diverse set of DGPs to be considered. Then, Monte Carlo simulations are used to generate series, each being fitted by all RS models. As many questions can arise from the broad motivation of this research, creating a very general and expandable setup and implementation is a goal in itself.

For the first part of the work, processing the Monte Carlo results starts with visualizing the generated series, understanding how each DGP ‘works’ and how RGP and SGP interact. With this in hand, the fit of the models can be visualized, checking which models captured the dynamics in which contexts. Then, more systematic regression analysis is done, explaining the performance of each estimated model by the DGP and model used, as well as interactions between the two, which can capture measures of mis-specification.

The second part builds on the theoretical framework. I hypothesize which characteristics of regimes can be relevant for model performance in each context, e.g., the conditional average for DGPs with intercept changes. Then, these distribution metrics are calculated for the Monte Carlo results. The processing follows the same steps as before, with initial visual diagnostics of whether the metrics indeed characterize the simulated series' regimes, and if the same pattern is found in the mis-specified models' results. Finally, regression analysis is done, now including the metrics as explanatory variables.

Aqui adicionarei o texto das perguntas mais concretas que foram pesquisadas, e os resultados.

The rest of this work is divided as follows: the general framework is presented in Section 3 and Section 4, while the specific implementation chosen is presented in Section 5 and Section 6. The exploratory analysis is done in Section 7, and the systematic analysis in Section 8. Finally, Section 9 concludes.

2 The Regime switching literature

The regime switching literature is vast and with many models' variations. It is important to map the models, the similarities and differences between them. Additionally, an important starting point is to discuss what is already known about their forecasting performance and the factors that influence it. Each is done in the sections below. Before doing so, I will better define the bounds of RS literature by discussing two closely related ones.

The first is the state-space (SS) literature, with its quintessential implementation in (Kalman, 1960). While RS and SS models have developed as somewhat independent fields, RS can be viewed as a subset of SS, where state (regime) variable and the observed series variable are modeled separately. This separation is central to the framework used in this paper. Bridges between the literatures include Switching State-Space Models and the seminal work by (Kim, 1994), which extends Hamilton's Markov-switching model to general state-space models.

The second is the structural break (SB) literature, the most relevant starting point being (Chow, 1960). Much of it is devoted to diagnosing breaks, which are indeed present in RS settings, with the non-constant parameters. However, SB models typically treat breaks as exogenous and non-recurring. Bridging the gap, works like (Bai; Perron, 1998) allow for multiple unknown breaks, which can be relevant for RS contexts, while (Chib, 1998) demonstrates that SBs can be formulated as Markov-switching processes that have only positive probability for staying in the initial regime and switching to the next, not for switching back.

2.1 Regime switching models

Two of the most essential aspects of the different RS approaches are: (i) if the latent regime variable is modeled in a deterministic or stochastic fashion, and (ii) if the changes between regimes are abrupt or smooth.

The most common deterministic models are the threshold-based ones, where some observable variable being above or below some threshold(s) is what determines the regime. The work of Howell Tong in (Tong, 1978) and (Tong; Lim, 1980) popularized the threshold autoregressive model, each regime having its own set of autoregressive parameters. Tong proposed that capturing smooth transitions between regimes would be important, and Timo Teräsvirta defined the smooth transition autoregressive model in (Teräsvirta; Anderson, 1992) and (Teräsvirta, 1994), where the distance between an observable variable and some threshold determines the continuous weight of each regime.

On the stochastic front, the Markov switching literature started with the seminal work of James Hamilton in (Hamilton, 1989) via the MSAR model, where the regime is governed by an unobservable Markov process – the probability of switching to another other regime is constant and depends only on the current regime. This implies in an geometric distribution for the amount of periods in a given regime's instance. The Markov switching smooth transition, as in (Elliott; Siu; Lau, 2018), model exist, but has more added complexity than the very natural jump from the TAR to the STAR model.

Moving forward, many variations on the regime variable modelling were created. The threshold variable have a delay or some transformation, it can be the series itself, an exogenous variable, or even a non-linear combination of variables (Chen; Liu; So, 2011). The probability distributions of MS models were extended to allow different distributions for the time spent in one regime, and dependence on more past values (Ferguson, 1980). The smooth transition function has several options, with common ones being the logistic and the exponential. The models were generalized to any number of regimes, with the STAR model being equivalent to a Neural Network, as described by (Medeiros, 2000).

Blurring the line between deterministic and stochastic models, (Chang; Choi; Park, 2017) uses threshold dynamics but adding an innovation that is dependent with the previous state’s innovation, and simplifies to a MS model when the threshold dynamic is exogenous and stationary. (Wu; Chen, 2007) creates a half-threshold half-random regime process. There are also unsupervised approaches of estimation, that make no assumption on the nature of the latent process, as in (Akioyamen; Tang; Hussien, 2020), where some clustering model can be used to identify regimes, and later the functional form can be estimated for each regime separately.

As I will note in this work, the functional form across regimes and the regime process itself are fairly independent, thus other variations arise from considering more complex functions than the autoregressive one. ARMA models have their RS counterparts (Brockwell; Liu; Tweedie, 1992). Not only the mean, but the variance can also be modelled: the ARCH/GARCH family, very relevant for finance, have their regime switching versions (Chen; Liu; So (2011)). More recently, models such as decision trees have been adapted to the regime switching context, as in (Adam; Ötting; Michels, 2024). Similarly, there are also models for vectors of times series.

General reviews on RS models include (Tan; Wu, 2025), (Potter, 2000), and (Hamilton, 2020), while (Chen; Liu; So, 2011) focuses on threshold models, (Dijk; Teräsvirta; Franses, 2002) in smooth transition, and (Song; Woźniak, 2021) in Markov switching. Note that RS models are considered in both frequentist and Bayesian frameworks, with the latter inheriting a lot from the SS models estimation literature.

2.2 Forecasting performance

There are many research topics in RS performance. I’ll focus on (i) important factors that relate to model selection and parametrization, to contextualize the decisions I made in this work; and (ii) comparisons between models, to contextualize the experiments I ran.

While RS models are frequently cited for their superior in-sample fit, which is useful for explaining historical phenomena, (Dacco; Satchell, 1999) noted that even minor errors in forecasting the future regime state can propagate through the non-linear structure, causing the overall prediction to perform worse than linear alternatives. Furthermore, standard metrics like mean squared error may be ill-suited for evaluating non-linear time series, potentially masking the utility of these models in capturing turning points or specific economic states.

A primary challenge in RS modeling is managing the trade-off between flexibility and overfitting. The most critical decision is on the number of regimes: too few can underfit, while too many might lead to overparametrization. Similarly, allowing, all parameters to switch can help capture complex dynamics and avoid mis-specification, but doing it when unneeded often dilutes out-of-sample power (Tan; Wu, 2025).

Each model also have their own specificities. For Markov Switching models, the estimation method is relevant: for example, EM algorithms have been noted to balance accuracy and speed in high-dimensional settings (Akbal, 2024). Moreover, the translation of soft posterior probabilities into hard regime labels affects accuracy, and different rules have different properties (Hall; Pagan, 2025). For deterministic models, the challenge lies in variable selection: identifying the correct threshold variable, delay parameter, or non-linear combination of variables remains a significant hurdle for effective specification.

Many papers compare the different RS models in many different contexts ((Clements; Krolzig, 1998), (Bierbrauer; Trück; Weron, 2004), (Pinson et al., 2008), (Janczura; Weron, 2010), (Elias; Wahab; Fang, 2014), (Chen; Bunn, 2014), (Panopoulou; Pantelidis, 2015), (Verne, 2021), (Aydın; Mermi, 2022)), and no single model is universally superior. The same context can present a different

“best” model depending on the focus (e.g. nowcasting, regime identification, portfolio performance, etc.) (Akbal, 2024).

TAR models are best employed when regime changes are triggered by a single, observable variable with rigid boundaries. They’ve shown effective for financial assets like gold prices and exchange rates, where transitions are fast rather than gradual (Aydın; Mermi, 2022). However, their reliance on observable triggers is a limitation: in contexts like offshore wind power, where fluctuations are driven by complex, non-observable states, TAR models fail to capture the underlying dynamics and significantly underperform compared to latent variable models (Pinson et al., 2008).

STAR models are theoretically appropriate for gradual economic adjustments but often face practical identification challenges. In many financial applications, the estimated smoothness parameter becomes so high that the model collapses into an abrupt threshold model, rendering the specific “smooth” specification inefficient (Aydın; Mermi, 2022). However, STAR models can outperform MS in macroeconomic contexts characterized by explosive volatility, such as GDP growth requiring the capture of “brutal” transitions typical of recession phases (Verne, 2021).

Markov-Switching Models (MS/MSAR) are the superior choice when regimes are driven by latent, multi-factor variables (e.g., market sentiment or meteorology) rather than a single observable index. But, with a more flexible regime framework, they have shown to be more sensitive to specification of the number of regimes (Bierbrauer; Trück; Weron, 2004), (Janczura; Weron, 2010).

3 Theoretical framework

In this section, I define the theoretical framework that guides the rest of this work. First, I define the general structure of RS DGPs, aligning all in a common mathematical representation, and relate the concepts of models and metrics to it. An important idea is the separation of the DGP into RGP and SGP.

3.1 The general regime switching DGP

Let $y_t \in \mathbb{R}$ denote the series of interest at time $t \in 1 : T^1$, $T \in \mathbb{N}$. Let $S \in \mathbb{N}$ denote the number of regimes. The *regime variable* is a vector $r_t \in \mathbb{R}^S$ of ‘weights’ for each regime, indexed by r_t^s , $s \in 1 : S$.

In this work, I consider only univariate series.

A DGP can be written in terms of a pair: *regime generating process* (RGP) and *series generating process* (SGP). This is essentially the separation between the state/system equation, and the output/measurement equation, in state space models. They are functions with parameters Θ_r and Θ_y , respectively, such that:

$$\begin{aligned} r_t &= \text{rgp}(y_{1:(t-1)}, r_{t-1}, t; \Theta_r) \\ y_t &= \text{sgp}(y_{1:(t-1)}, r_t, t; \Theta_y) \\ &= \text{sgp}(y_{1:(t-1)}, \text{rgp}(y_{1:(t-1)}, r_{t-1}, t; \Theta_r), t; \Theta_y) \end{aligned} \quad (1)$$

Without loss of generality, I restrict the regime weights to be non-negative and sum to one, i.e. $r_t \in [0, 1]^S$ and $\sum_{s=1}^S r_t^s = 1$. Notably, the number of regimes S is a parameter in Θ_r , and Θ_y is actually a set of different parameters for each regime, each indexed by Θ_y^s . This means that the SGP can be written as:

$$\text{sgp}(y_{1:(t-1)}, r_t, t; \Theta_y) = \sum_{s=1}^S f_{\text{sgp}}(y_{1:(t-1)}, t; \Theta_y^s) \cdot r_t^s \quad (2)$$

Note how each regime is weighted by the regime variable r_t^s . In the simplest case, this weight is binary ($r_t \in \{0, 1\}^S$) – only one of the regimes is ‘turned on’, and all the others are ‘turned off’. However, in some models, such as Smooth Transition, the weights can be continuous, with different regimes being partially ‘on’ at the same time.

Furthermore, Θ_y could encode different functional forms for each regime. As this is not common, I refer to f_{sgp} as the *SGP functional form*, or simply *SGP*, and the set of parameters Θ_y as the *regime nature*, as they define what actually changes between each regime. Figure 1 illustrates this structure.

Figure 1: The general RS DGP structure.

$$\begin{array}{c} \text{DGP} = \underbrace{\sum_{s=1}^S \text{sgp}(\cdot; \Theta_y^s) \cdot r_t^s}_{\text{SGP}} \quad \& \quad \underbrace{r_t = \text{rgp}(\cdot; \Theta_r)}_{\text{RGP}} \\ \underbrace{\hspace{10em}}_{\text{functional form} \quad \& \quad \text{regime nature}} \\ \underbrace{\hspace{10em}}_{f_{\text{sgp}} \quad \& \quad \{\Theta_y^1, \dots, \Theta_y^s\}} \end{array}$$

¹Let $a : b := \{a, a + 1, \dots, b\}$ for $a \leq b \in \mathbb{Z}$, and $y_{a:b} := \{y_a, \dots, y_b\}$.

To construct a diverse set of DGPs, I combine different RGPs, SGPs, and regime natures. One of the challenges of this work is to choose a comprehensible set of these elements, and analyze their differences in a systematic but manageable way.

In the notation above I omitted the error term. For our purposes, it is more useful to write the DGP as a function that receives a sequence of random errors², and returns the series and the regimes:

$$(y_{1:T}, r_{1:T}) = \text{dgp}(\varepsilon_{1:T}; \Theta_r, \Theta_y) \quad (3)$$

Consider the notation shorthand $y := y_{1:T}$, and similarly for other variables, used for the rest of this work.

Let the set of considered DGPs be P (for ‘processes’). These are present in the literature, as discussed in Section 2, and will be defined in Section 5.

3.2 Models

Consider a model mod as a function with parameters Θ_m that generates the fitted values and H -step ahead predictions of the series and regimes. The model can also return a set $\hat{\pi}$ of general metadata, e.g. the estimated parameters.

$$(\hat{y}, \hat{r}, \hat{\pi}) = \text{mod}(y_{1:(T-H)}; \Theta_m) \quad (4)$$

Notably, the number of estimated regimes $\hat{S}$ is a parameter in Θ_m , which may or may not be equal to S .

Let the set of models be M (for ‘models’). Also present in the literature, they will be defined in Section 5.1.

3.3 Metrics

A regime-conditional (RC) metric c is function that receives a vector of series and a vector of regimes, and returns a sequence with one value for each regime. They are used to characterize the distribution of y_t or (y_t, y_{t-j}) within each regime.

$$c : (y, r) \mapsto \mathbb{R}^S \quad (5)$$

An example is the function that returns, for each regime s , the mean of the series weighted by r_t^s . This can be done for many common metrics, and is equivalent to mapping (y, r) to the S sets R_s of regimes’ observations³, then applying the metric to each set. The benefit of the first approach is that it is more general, allowing for non-binary – i.e. smooth transition – regimes.

For the joint distribution (y_t, y_{t-j}) , the metrics are more complex, as they must consider only the windows $(y_t, \dots, y_{t-j})$ fully contained in the same regime instance. This is further described in Section A.2.

In any case, RC metrics possibly lump together observations from different time windows. For them to describe a well-defined regime distribution, it is required that the series be stationary within each regime. This will impose restrictions on the DGPs that this work will consider.

²This is a simplification, assuming the same error distribution across regimes.

³ $R_s := \{y_t : r_t^s = \max\{r_t\}\}$.

3.3.1 Within-regime stationarity

For the distribution of a regime to be well defined, all datapoints within it must be drawn from the same distribution. Formally, *within-regime strong stationarity* requires, for all regimes $s \in S$, a CDF F_s such that:

$$F_s(y_{t_1}, \dots, y_{t_n}) = F_s(y_{t_1+j}, \dots, y_{t_n+j}) \quad \forall \{y_{t_1}, \dots, y_{t_n}\}, \{y_{t_1+j}, \dots, y_{t_n+j}\} \subset R_s, \quad \forall j \in \mathbb{N}_* \quad (6)$$

As we intent to characterize the distributions with specific metrics, weaker assumptions can be made. If we restrict ourselves to the moments of the (joint) distribution, we can require the weak version. Formally, *within-regime weakly stationarity* requires⁴, for all $s \in S$, that:

$$\begin{aligned} E[y_t] &= E[y_{t'}] & \forall y_t, y_{t'} \in R_s \\ \text{Var}[y_t] &= \text{Var}[y_{t'}] & \forall y_t, y_{t'} \in R_s \\ \text{Cov}[y_t, y_{t-j}] &= \text{Cov}[y_{t'}, y_{t'-j}] & \forall \{y_t, \dots, y_{t-j}\}, \{y_{t'}, \dots, y_{t'-j}\} \subset R_s, \\ & & \forall j \in \mathbb{N}_* \end{aligned} \quad (7)$$

Processes that have a non-binary r_t , i.e. smooth transitions don't have truly separated regimes, and thus, generally don't satisfy the conditions above. Thus, the metrics cannot be interpreted as e.g. "the mean of all datapoints in an regime". Still, their information might be useful, as will be studied in this work.

3.3.2 Aspects of RC metrics usage

How the metrics will be used is going to be discussed later, but for now, it is important to note there are two aspects of their use. First is whether to use the whole sequence of values for each s , or to condensate it into a single value of dispersion across regimes. For example, the average pairwise distance between the RC means, a single value that describes how distant are the levels of the regimes. This is equivalent to composing an dispersion function d :

$$d \circ c : (y, r) \mapsto \mathbb{R}^S \mapsto \mathbb{R} \quad (8)$$

Second, which series to use, the true or estimated ones. One can use the true values (y, r) and get the characteristics of the true DGP, or the estimated values $(\hat{y}, \hat{r})$ and get the characteristics of the estimated model⁵. Another option is to calculate the difference between the former and the latter⁶. Another option is to calculate the metric of the difference $(y - \hat{y}, r)$ or $(y - \hat{y}, \hat{r})$.

This framework allows for mixing and matching these options, each being useful to answer different questions. In this work, I focus on the estimated series, as they are the only thing available to the econometrician in practice, and in using the dispersion of RC metrics, as it is more comparable across DGPs and models.

⁴The ACF condition should be read as "j'th autocorrelations between time-points within the same regime instance should always be equal", not as "j'th autocorrelations of every time-point in the same regime instance should always be equal", although the latter is true for AR processes with order higher than j .

⁵Note that the value of S and $\hat{S}$ can be different, and thus, so the dimension of the metric's output.

⁶This is only possible if $S = \hat{S}$ and there is an unambiguous way to match the estimated and true regimes.

Não sei se foi uma tangente muito grande falar dessas outras opções que não vou usar, ainda mais porque pode não estar claro pro leitor qual tipo de pergunta cada ajuda a responder. Eu gostaria de informar essa flexibilidade do framework, mas talvez valha mais colocar isso num apêndice.

Let the set of metrics $d \circ c$ be C (for ‘criteria’). These will be defined in Section 5.4, but are mostly based on the moments of y_t and of the pair (y_t, y_{t-j}) , $j \in \mathbb{N}$, and the performance metrics for the dependent variable.

4 Simulation framework

One of the partial goals of this work was to create the theoretical framework described in the last section in a very general and expandable way, that easily allows for different exercises, even if they are not considered here. Similarly, the simulation structure was designed to follow the same concept.

There are the following steps to perform the simulations:

1. Generate random errors for all the DGPs.
2. For each DGP and simulation, generate (y, r) .
3. For each DGP, simulation, and model, obtain $(\hat{y}, \hat{r})$.
4. For each DGP, simulation, and model, compute each metric.
5. Aggregate the metrics, performance information, and DGP and model descriptors into a dataset.

4.1 Forecast horizon

For the forecast performance, I focus on 1-step ahead predictions. It would be interesting to expand that, be it with locally-projected models or not.

To obtain more than one prediction per simulation, I simulate a $T - H$ -long series, and obtain H predictions. There are two possible approaches:

1. For each iteration $h \in 1 : H$, the model is estimated with the window $h : (T - H + h - 1)$, and generates $\hat{y}_{T-H+h}$.
2. The model is estimated once with the window $1 : (T - H)$, then for each h , $\hat{y}_{T-H+h}$ is generated using $y_{1:(T-H+h-1)}$.

The second approach is computationally cheaper, allowing for more simulations and DGPs to be considered. It is the one used in this work, but note that it is less accurate to what would be done in practice, as econometricians often re-estimate their models with new data.

4.2 Simulation parameters

The parameters of the simulation are as follows:

- Number of simulations: I . Its main effect is on the the precision of the results, and diversity of series.
- Forecast horizon: H predictions of 1-step ahead values. Also affects the precision of the results, but does not change the diversity of series.
- Total number of observations: T . Its main effect is on the ability of the models to learn the dynamics and separate the regimes. Results for higher T 's are more relevant for contexts with a lot of data, such as high-frequency financial data, while lower T 's are more relevant for contexts with less data, such as macroeconomic data.
- Burn-in period: B . Its main effect is on reducing the dependence of the initial values, but with stationary processes, this is not too problematic.

Let $i \in 1 : I$, $I \in \mathbb{N}$ be the simulation index.

4.3 Simulating Series

I will only consider DGPs have the same error distribution – but note that a DGP can have a volatility parameter multiplying its error. For each DGP, indexed by $p \in 1 : |P|$, there are I random error vectors created, each of size T . Let E denote the set of all errors. Let $E_{p,i}$ denote the vector of errors generated for the p -th DGP and the i -th simulation. Similar indexing definitions will be used for similar collections throughout this document.

Let Y and R denote the sets of generated series and regime variables. They are computed given $E_{p,s}$:

```

1: Initialize  $Y$  and  $R$ 
2: for  $p$  in  $1 : |P|$  do
3:   for  $i = 1$  to  $I$  do
4:      $Y_{p,i}, R_{p,i} \leftarrow P_p(E_{p,i})$ 
5:   end for
6: end for

```

4.4 Estimating Models

Now, for each simulation, I estimate each model, generating the sets $\hat{Y}$, $\hat{R}$, and $\hat{\Pi}$. The models are trained using only $y_{(B+1):(T-H)}$, to avoid the burn-in period and leave space for the forecast horizon.

The nesting order is the same as above, for consistency, but with an additional inner loop for the model estimation.

```

1: Initialize  $\hat{Y}$  and  $\hat{R}$ 
2: for  $p$  in  $1 : |P|$  do
3:   for  $i$  in  $1 : I$  do
4:     for  $m$  in  $1 : |M|$  do
5:        $\hat{Y}_{p,i,m}, \hat{R}_{p,i,m}, \hat{\Pi}_{p,i,m} \leftarrow M_m(Y_{p,i}, R_{p,i})$ 
6:     end for
7:   end for
8: end for

```

4.5 Calculating Metrics

Then, for each model, the dispersion of the RC metrics are calculated and stored as columns of a dataset D . Each row of D is identified by (p, i, m) .

```

1: Initialize  $D$ 
2: for  $(p, i, m)$  in  $(1 : |P|) \times (1 : I) \times (1 : |M|)$  do
3:   for  $c$  in  $1 : |C|$  do
4:      $D_{(p,i,m), c} \leftarrow C_c(\hat{Y}_{p,i,m}, \hat{R}_{p,i,m})$ 
5:   end for
6:    $\hat{\Pi}_{p,i,m}$  is appended to  $D_{p,i,m}$ .
7:   Categorical variables  $(p, i, m)$  are appended to  $D_{p,i,m}$ .
8: end for

```

Recall the discussion in Section 3.3 about the two different aspects of RC metrics usage. With different choices regarding the usage of true or estimated series, the function C_c could receive different inputs. Additionally, the function could return the whole sequence of RC metrics, not a single value, then, each row would be identified by (p, i, m, s) .

The dataset D is already in a friendly format for analyzing the relationship between the performance of each observation and the characteristics of the regimes, as well of considering stratifications by DGP and model.

5 Considered DGPs, models, and metrics

5.1 Considered SGPs

The functional form of the SGP could be important in its interaction with the other ingredients of the DGP. Additionally, some topics are interested in specific SGPs, such as conditional volatility in finance and GARCH models. For now, however, this does not seem to be the main point of interest. I will consider only an $AR(1)$ process, for its simplicity, popularity, and ease of estimation.

As discussed, it is useful to consider only within-regime weakly stationarity, even though many interesting DGPs are non-stationary. This restricts the absolute value of the $AR(1)$ parameter to 1. The only SGP functional form considered is the following:

$$\begin{aligned} f_{\text{sgp}}(\cdot; (\mu^s, \rho_1^s, \sigma^s)) &= \mu^s + \rho_1^s y_{t-1} + \sigma^s \cdot \varepsilon_t \\ \varepsilon_t &\sim \mathcal{N}(0, 1) \\ |\rho_1^s| &< 1, \quad \sigma^s > 0 \end{aligned} \tag{SGP-AR(1)}$$

Several others SGP's could be considered, such as ones with transformations of y_t as regressors, non-linear regression forms, or even decision trees, as in the common model Markov-switching Random Forest. Still, the $AR(1)$ is an essential building block, and its simplicity helps isolating the effects of the other ingredients.

5.2 Considered RGPs and models

The next 'ingredient' is the RGP. I will consider the options Self-Exciting Threshold (SET), Smooth-Transition (ST), and Markov-Switching (MS). Structural Break (SB) is included to study how RS models perform in the case of breaks without reoccurring regimes.

Each of these RGPs have empirical model counterparts, which are also considered. There is an additional model with an unsupervised approach where the regimes are defined by some clustering technique and each regimes' AR is estimated independently afterwards (Clustering + AR, CAR). Finally, a non-RS Random Forest (RF) model is included as a benchmark.

The formal definition of each RGP/model is presented in the Section A.1, first the RGP hypothesis, then the empirical model's estimation strategy.

For all RGPs, it is considered an option with equally likely regimes, and an asymmetric variation.

- **Structural Breaks:**

- A single break at $T/2$, and a single break at $2T/3$.

- **Self Exciting Threshold:**

- Fixed parameters: switching based on y_{t-1} . Different lags are often specific to timing-related issues, and not considered here.
- A single threshold at 0, and a single threshold at 0.5.

- **Smooth Transition:**

- Fixed parameters: switching based on y_{t-1} , logistic's CDF as transition function.
- A single threshold at 0, and a single threshold at 0.5.

- **Markov Switching:**

- Symmetric matrix, high persistence ($P(s|s) = 0.9$), symmetric matrix, low persistence ($P(s|s) = 0.6$).
- Asymmetric matrix, high persistence ($P(1|1) = 0.9$, $P(1|2) = 0.7$), asymmetric matrix, low persistence ($P(1|1) = 0.8$, $P(1|2) = 0.6$).

Tem várias outras parametrizações já feitas no código e com o texto escrito, mas deixei só essas aqui caso algo mude. Alguma sugestão de formatação melhor do que a atual?

For the models, most hyperparameters are as follows:

- All the coefficients are assumed to change across regimes, as this is common assumption, especially in the face of possible mis-specification.
- The number of regimes $\hat{S}$ is fixed, not estimated. Models are estimated with 2 regimes.
- The values of model-specific hyperparameters are the same as the related RGP's values.

5.3 Considered regime natures

The following regime natures are considered, each representing a different way in which the SGP parameters change across regimes:

- **Mean (μ) change:**
 - Small difference: ($\mu^1 = 0$, $\mu^2 = 0.5$)
 - Large difference: ($\mu^1 = 0$, $\mu^2 = 2$)
- **Persistence (ρ_1) change:**
 - Small difference: ($\rho_1^1 = 0.6$, $\rho_1^2 = 0.4$)
 - Large difference: ($\rho_1^1 = 0.9$, $\rho_1^2 = 0.1$)
- **Volatility (σ) change:**
 - Small difference: ($\sigma^1 = 1$, $\sigma^2 = 2$)
 - Large difference: ($\sigma^1 = 1$, $\sigma^2 = 4$)

Idem, existem outras opções. Alguma sugestão de formatação melhor do que a atual?

Note that the regimes are always ordered increasingly by the parameter of interest. In general, the large vs. small differences will be interesting to analyze in relation to each other. To compare different types of changes, only the large differences will be considered, for simplicity.

5.4 Considered metrics

The goal with RC metrics is to capture the change in the series characteristics across regimes. One important option is the estimated parameters of the model for each regime, e.g. $(\hat{\rho}_s)_{s \in \hat{S}}$, $(\hat{\mu}_s)_{s \in \hat{S}}$, etc. One might think that this would outshine all other metrics, but in more complex cases where more than one parameter changes, this becomes less useful. More general metrics generate benefits from their abstraction over the DGP. Additionally, in simple SGPs, there often is a metric that is directly connected to changes in parameters, such as conditional average for changes in intercept.

In this work, I focus on the moments of the distribution of y_t and (y_t, y_{t-j}) . Specifically, the RC metrics considered are the RC mean, RC standard deviation, and RC autocorrelation of lag 1. Higher lags could be considered, but in the simple $AR(1)$ context this would bring little additional information.

Talvez o 3ro e 4to momentos sejam interessantes, especialmente o 3ro porque alguns DGPs geram séries assimétricas.

As stated before, the RC mean and RC SD are simply the mean and SD of each set R_s . The autocorrelation is similar, but must be calculated separately for each concurrent set of observations in R_s . The formal definitions are stated in the Section [A.2](#).

As the focus is on the dispersion of RC metrics, two important measures to consider are the standard deviation and the average pairwise absolute difference. For only two regimes, they are very similar and the absolute difference is more intuitive. All the metrics are composed such as all $d \circ c \in C$ return a single real value, and $d(x) = |x_1 - x_2|$.

There are some possible expansions on this work's metrics calculation. One is to use non-standard weights for the empirical moments, giving more importance to observations near the edges of regimes' instances. Another is to use a cluster separation measure, such as the silhouette score, instead of a simple absolute distance between the RC metrics. Finally, one can use distribution distance metrics, such as the Earth Mover's Distance, on the empirical distribution of each regime.

The list of considered metrics is as below:

- First moment: RC mean $\hat{\mu}(y|S)$.
- Second moment: RC standard deviation $\hat{\sigma}(y|S)$.
- First autocorrelation: RC 1st autocorrelation $\hat{\rho}_1(y|S)$.

5.4.1 Performance and RGP metrics

The performance metrics considered are R^2 for fit performance, and RMSE and MAPE for forecasting performance.

Other metrics pertaining the RGP will be included as controls in the regression analysis: the average duration and number of instances of a regime, and transition probabilities.

SGP	RGP	R1		R2		ANOVA
		Mean	SD	Mean	SD	P-value
μ	Markov (symm., high persist.)	0.30	0.27	3.60	0.29	0.0e+00***
μ	S-break (at 1/2)	-0.01	0.28	3.92	0.24	0.0e+00***
μ	Threshold (x, at 0)	0.17	0.79	3.96	0.19	0.0e+00***
μ	LSTAR (at 0)	1.14	0.11	0.98	0.23	6.2e-43***
ρ	Markov (symm., high persist.)	0.32	0.13	0.48	0.14	2.9e-70***
ρ	S-break (at 1/2)	0.36	0.13	0.55	0.11	1.0e-105***
ρ	Threshold (x, at 0)	0.13	0.14	0.26	0.13	4.7e-49***
ρ	LSTAR (at 0)	0.15	0.14	0.24	0.14	1.5e-27***
σ	Markov (symm., high persist.)	1.35	0.21	4.40	0.55	0.0e+00***
σ	S-break (at 1/2)	1.13	0.13	4.50	0.48	0.0e+00***
σ	Threshold (x, at 0)	1.27	0.13	4.16	0.49	0.0e+00***
σ	LSTAR (at 0)	1.77	0.16	3.66	0.47	0.0e+00***

6 Simulation implementation

6.1 Implementation

The implementation of the simulations, as well as their analysis in the next sections, is done with the R programming language, and the code can be found in [this paper's repository](#). The code is highly modular and fully reproducible, following the intent of setting up an expandable framework.

The error sequences were generated in parallel, using [TRNG](#).

6.2 Diagnostics

The errors should be i.i.d. Gaussian with mean 0 and should not present any pattern, especially across the parallelization structure. This is guaranteed by the TRNG library, and is checked in Section [B.1](#).

On top of visualizing the series, to further check for problems in the series generations, the regime-conditional and unconditional moments are estimated and tested against their true values. Additionally, the ANOVA test of equal moment on all regimes is done. The regime-conditional true values are calculated as the standard $AR(1)$ moments. There is only an analytical formula for the unconditional moments of the SB and MS RGP, calculated via iterated expectations.

Table **TODO** shows the results. Each group of lines corresponds to the moments of a DGP. The first two columns relate to the values conditional in regime 1 and 2, the third column gives the unconditional values. Each cell has the value of the moment, and in brackets the p-value of the null hypothesis that the moment is equal to its true value. The last column shows the ANOVA p-value.

Os valores finais dos diagnósticos podem mudar. Coloco aqui uma tabela de placeholder.

We can see that the results are generally consistent with the expectations. The table is also useful to better understand the difference between the DGPs. Some useful analyses to be done are: as expected, the first moment is often informative about regime natures with changes in intercept, and similarly for the other moments; the RGP and regime nature interact in a non-trivial way, e.g. the TAR regime with higher intercept generates an asymmetric process, as its 'hard to escape' that regime; How each DGP differs in terms of separation between their regimes, but more on that in the next section.

To check the models, it is expected that models with the same RGP assumption as the DGP return similar moments, so a similar analysis as above is done in the Section [B.2](#). The results are generally consistent with the expectations.

As a final sanity check, the frequency of improbable events were annotated: TODO% of the datapoints generated were 10 standard deviations away from the relevant mean; and TODO% of the predictions were 10 standard deviations away from the true value.

6.3 Regime separation

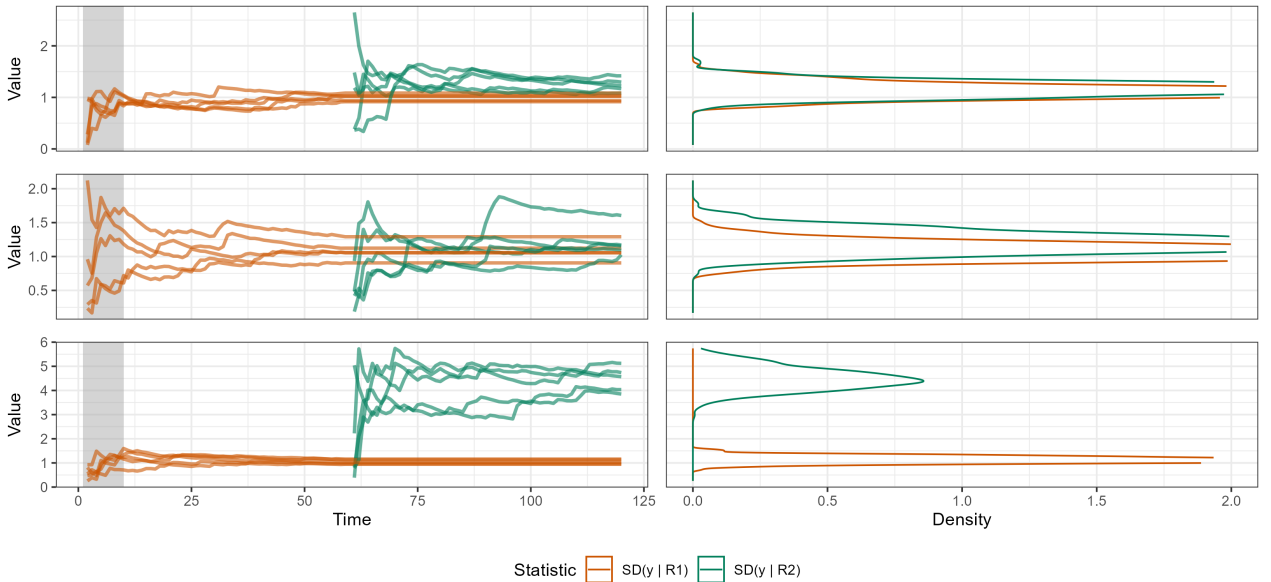
There are several aspects in which each DGP's generated series differ, and how the hyperparameters contribute to that difference. I propose that one of the most interesting aspects to analyze is the degree of separation of regimes, in terms of the metrics. This could be seen in table TODO, but I argue a better visualization is to graph how the separation progresses across the sample windows $1 : 2$, $1 : 3$, ..., $1 : T$.

By graphing the latest time-point considered in the calculation of the metrics on the x-axis, and the value of the metric on the y-axis, we can see how the separation evolves across sample size. This is useful because the sample size is one of the most important factors for the models to learn how to separate the regimes.

Item. Coloco aqui a análise antiga só para referência.

Figure 2 shows the metrics for the SB-AR(1) model, with a break at the middle. The left-hand side shows the metrics calculated with the data up to the time on the x-axis, yielding a ‘rolling’ metric that can be used to analyze convergence. The right-hand side shows the distribution of the metrics calculated with the full series, for all simulations. We can see how the second regime only has information starting from $T/2$. Each row represents a different regime nature, thus a different conditional metric (mean, ACF(1), and SD, respectively).

Figure 2: SGP metrics of SB-AR(1) model

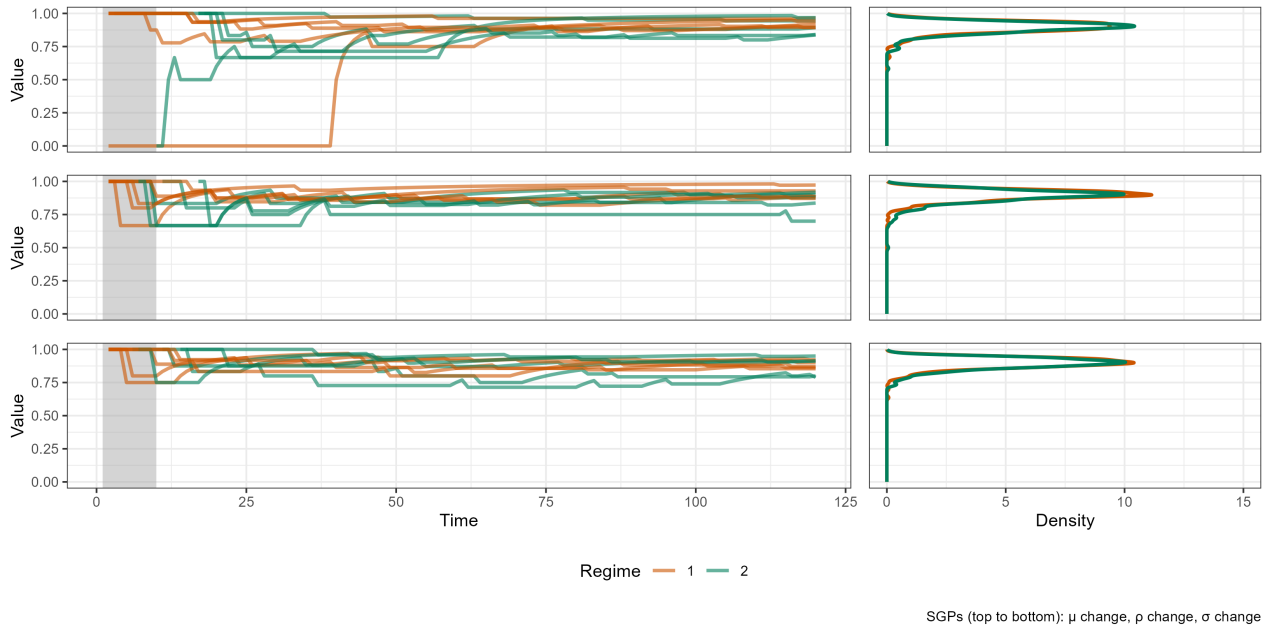


SGPs (top to bottom): μ change, ρ change, σ change

Figure 3 shows a metric for the RGP itself, for the MS-AR(1). Specifically, the empirical (non-)transition probability for each regime.

In a more systematic way, the table below shows the average and standard deviation of the metrics and DGPs, and the columns ANOVA present the p-values of the null hypothesis that the metrics

Figure 3: RGP metrics of MS-AR(1) model



do not vary across regimes. Note that the power of this test is directly related to the number I of simulations.

7 Exploratory Analysis

Creio que explorar tudo que contei na motivação poderia ser sua própria seção. Gera alguns resultados ‘fatos estilizados’ sobre os DGPs e como os modelos interagem com cada, mas também gera insumos pra seção seguinte. Ainda assim, não está desenvolvida. Coloquei apenas alguns exemplos de gráficos e interpretações, pra ter uma ideia do que poderia ser feito.

Before doing the systematic analysis and focusing on the results that could generate practical recommendations, it is important to explore the data, learn how each DGP behaves, and how the models interact with them. This will yield facts relevant in their own right, but also motivate the modeling decisions for the systematic analysis.

A benefit of having only one parameter changing in the DGP at a time is that regimes can be ordered by it, even the estimated ones. As noted in TODO, regimes are always ordered increasingly by the parameter of interest, and the same is done for the estimated regimes in the figures of this section.

7.1 Series

Processing the Monte Carlo results starts with visualizing the generated series, understanding how each DGP ‘works’ and how RGP and SGP interact.

7.1.1 Values and Distribution

Figure 4 show the series for the MS-AR(1) model, with a symmetric high-persistence transition matrix. The left-hand side shows a single simulated series, while the right-hand side shows the distribution of all simulations. Each row represents a different regime nature, with only μ changing, only ρ_1 changing, and only σ changing, respectively. The grey area is the burn-in period.

We can see how the high persistence of regimes is indeed present in the data, with long periods in each regime. The change in intercept and volatility are clear, while the higher ρ_1 conditions a more volatile regime, as the past errors have a bigger impact.

Figure 5 shows the series for the SET-AR(1) model, with a threshold at 0. Here, the interaction between RGP and the regime nature is evident: the higher μ makes the series stray away from the threshold and very likely stay in regime 2. The higher volatility and ρ_1 , which happen only when the series is above 0, also end up conditioning a higher level for the series in these regimes.

Many other observations could be made. For now, the most important information to note is how the RGP and regime nature interact, creating different, non-obvious patterns in the series.

7.2 Models

The processing follows the same steps as before, with initial visual diagnostics of whether the metrics indeed characterize the simulated series’ regimes, and if the same pattern is found in the mis-specified models’ results.

Figure 4: Values of MS-AR(1) model

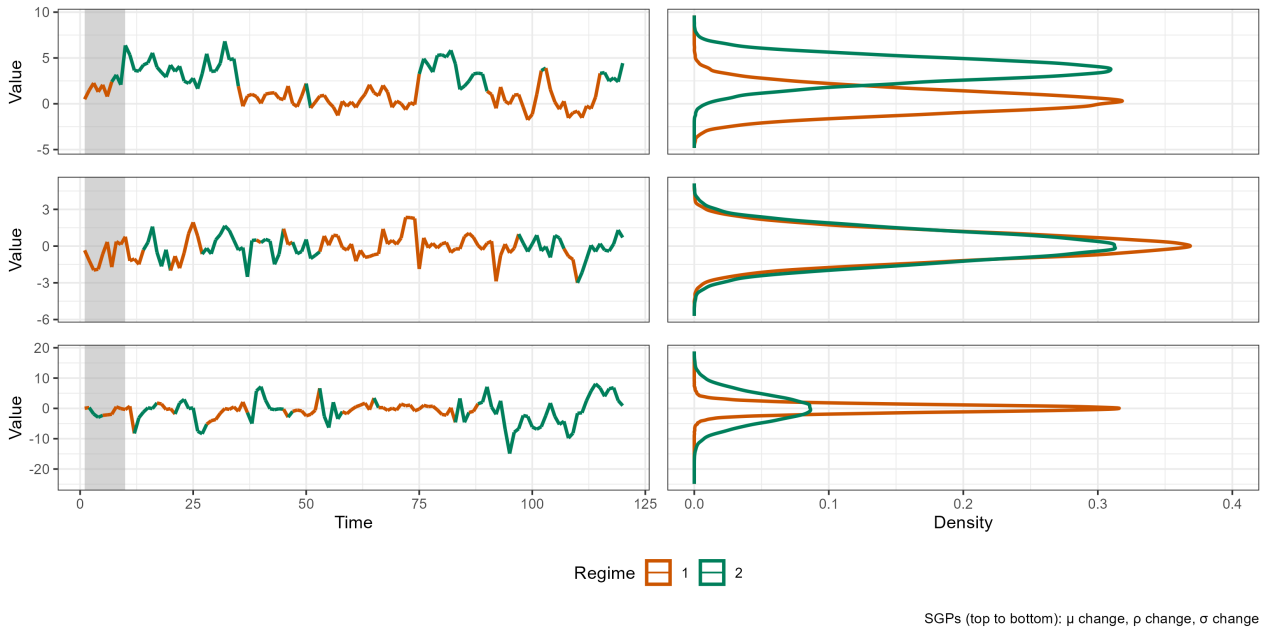
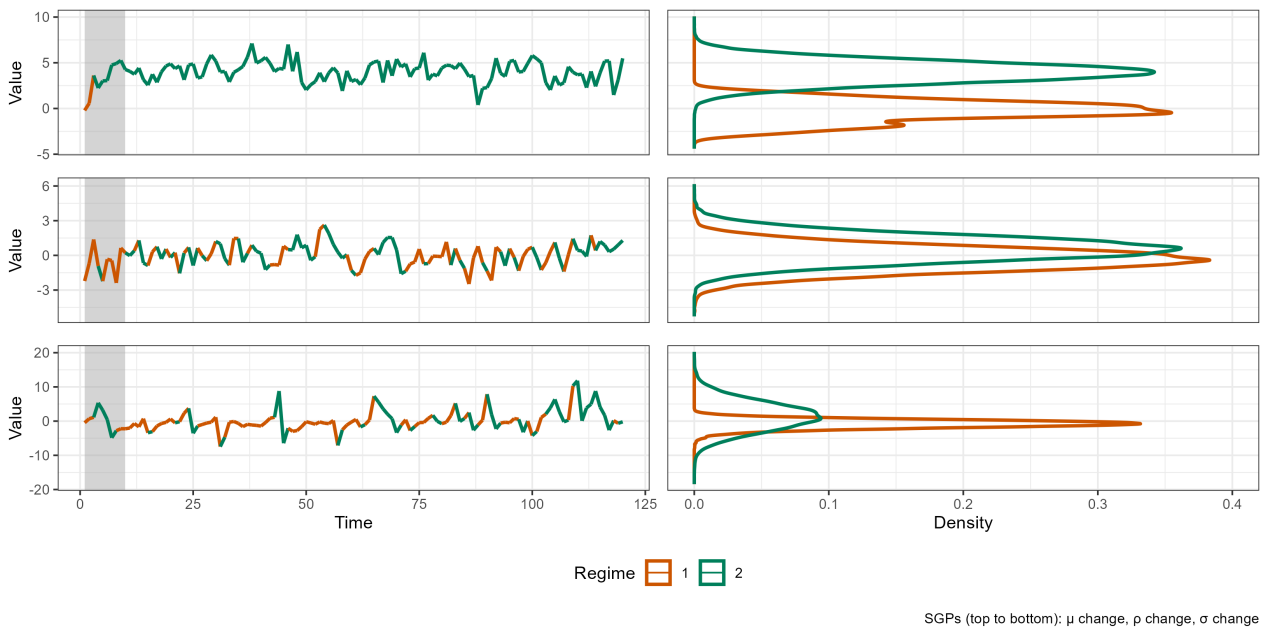


Figure 5: Values of SET-AR(1) model

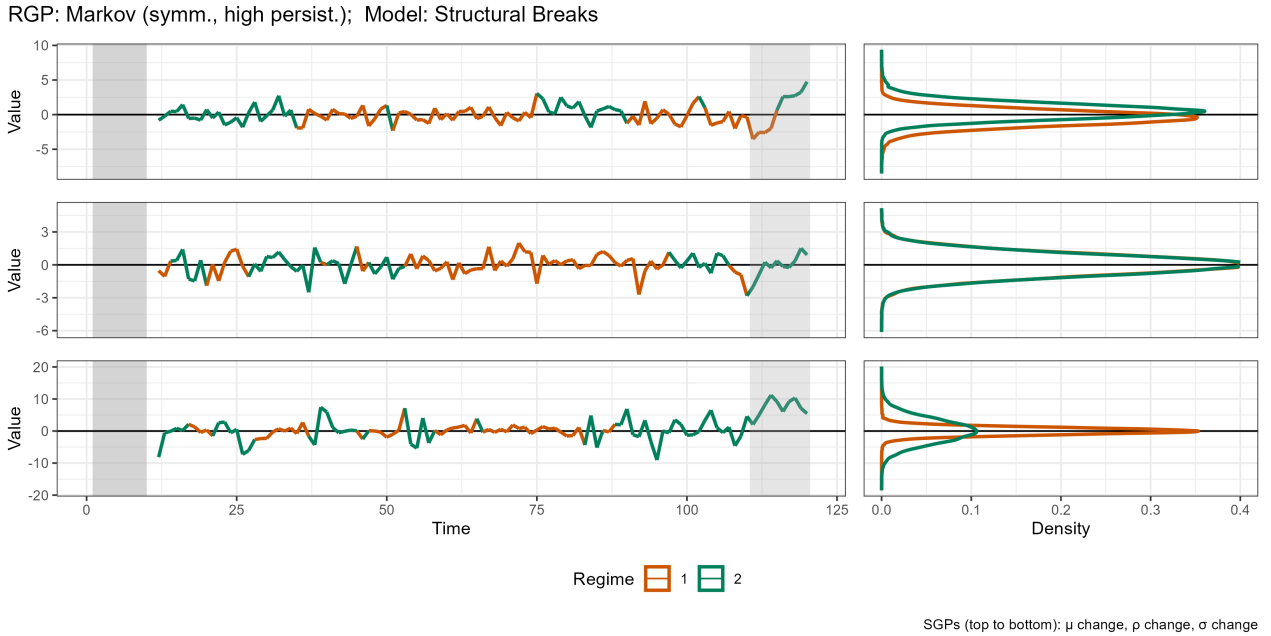


SGP	RGP	R1		R2		ANOVA
		Mean	SD	Mean	SD	P-value
μ	Markov (symm., high persist.)	0.65	18.32	0.11	0.21	5.0e-01
μ	S-break (at 1/2)	-0.14	0.09	0.03	0.10	3.2e-127***
μ	Threshold (x, at 0)	-0.07	0.15	-0.03	0.14	4.1e-05***
μ	LSTAR (at 0)	-0.11	0.11	0.08	0.13	5.1e-110***
ρ	Markov (symm., high persist.)	-0.01	0.14	-0.00	0.16	6.4e-01
ρ	S-break (at 1/2)	-0.01	0.14	-0.01	0.39	5.9e-01
ρ	Threshold (x, at 0)	0.18	4.32	1.26	28.20	4.0e-01
ρ	LSTAR (at 0)	-0.01	0.14	0.02	0.15	7.1e-03***
σ	Markov (symm., high persist.)	-0.07	0.28	-0.09	3.01	8.7e-01
σ	Threshold (x, at 0)	-0.37	0.19	0.37	0.17	0.0e+00***
σ	LSTAR (at 0)	0.32	0.13	-0.37	0.23	0.0e+00***

7.2.1 Residuals

The first question is about the model's fit. Figure 6 shows the residuals and their distribution for the MS-AR(1) model, estimated on top of an SB RGP. While figure Figure 6 indicates the estimated regimes in the colors, figure Figure 7 indicates whether the regime was correct or not.

Figure 6: Residuals of MS-AR(1) estimating a SB-AR(1)



It appears that the estimated regime relates more to the residual level than its actual correctness. The differences in average and volatility are expected, and no difference can be seen in the autocorrelation change.

Again, in a more systematic way, the table below shows the average and standard deviation of the residuals across estimated regimes.

Sinto que os gráficos dos resíduos ajudam a resumir a coisa, em vez de mostrar os pares fit-real.

7.2.2 Coefficients

The next step is to analyze if the models are able to capture the coefficients and their difference across regimes. Note that this might not be a necessary condition for a good approximation.

Figure 7: Residuals of MS-AR(1) estimating a SB-AR(1)

RGP: Markov (symm., high persist.); Model: Structural Breaks

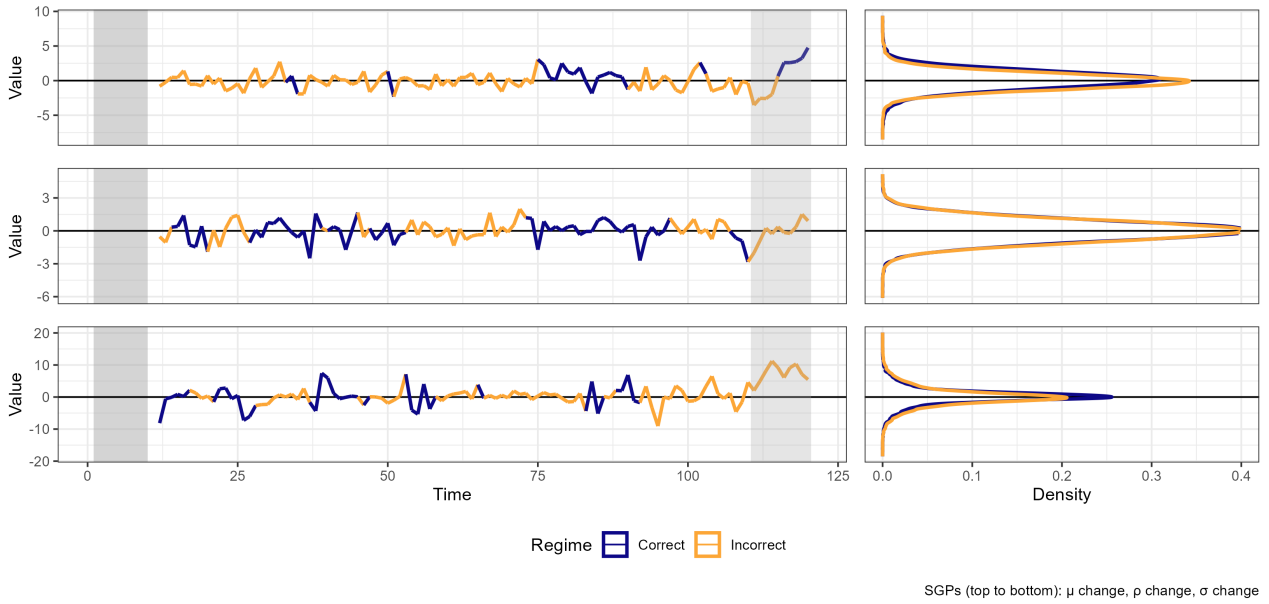


Figure 8 follows the same model from before and shows the distribution of each estimated coefficient (μ and ρ_1), while the dotted lines give the true values. Now the rows are separated by the big-and-small changes in each regime nature.

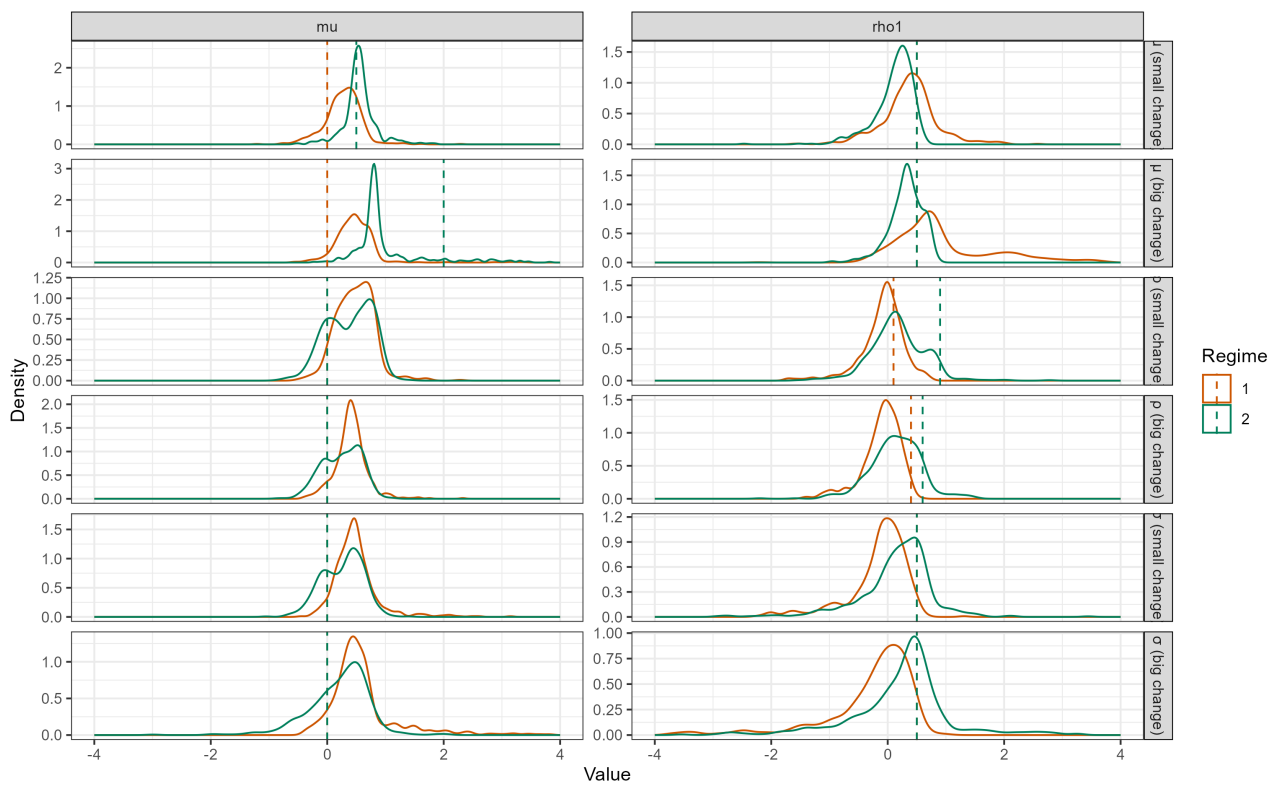
It is important to note both how the coefficients that *should* change do (or don't), and how the coefficients that *shouldn't* change might be compensating for the mis-specification.

Eventualmente gostaria de colocar uma terceira coluna com a volatilidade estimada. Vou ajeitar o eixo x também.

7.2.3 Metrics

Similar à seção de métricas dos DGPs, seria possível fazer análises similares para os modelos. Ver se as mesmas métricas seguem caracterizando os regimes ou não, etc.

Figure 8: Coefficients of MS-AR(1) estimating a SB-AR(1)



8 Systematic Analysis

Eu já rodei uma versão inicial dessas regressões, só não coloquei as tabelas aqui, dado que ainda precisamos filtrar o que vai ser realmente mantido. Similarmente, também criei scatterplots para as regressões. As interpretações também são bem iniciais.

As regressões abaixo podem ser rodadas a nível de regime também (p, i, m, s) . A interpretação só muda para “em qual regime o modelo costuma errar mais”.

8.1 Stylized Facts about DGPs

In the first exercise, I run regressions with fixed effects for DGP and model, separate the model effects into RGP, SGP, and regime nature effects, and analyze these effects. Also, I study the sensitivity to mis-specification, via an indicator variable $\mathbb{1}(p = m)$ or with a full interaction $m \times p$.

An important placebo test is to check if the simulation index i has no effect, via the regression below:

$$rmse_{p,i,m} = \beta_0 + \beta_1 i + \varepsilon_{p,i,m} \quad (9)$$

We in fact find no effect, as expected.

Next, we turn to the DGPs. Consider the categorical variables $\text{rgp}_{p,i}$, i.e., vector of dummies, that indicates which RGP was used, and similar definitions for the RGP and the models. The regression below analyzes the fixed effects of RGP, SGP, and interactions between them, as was shown to be relevant in the exploratory analysis.

$$rmse_{p,i,m} = \beta_0 + \beta_1 \text{rgp}_{p,i} + \beta_2 \text{sgp}_{p,i} + \beta_3 \text{rgp}_{p,i} \cdot \text{sgp}_{p,i} + \varepsilon_{p,i,m} \quad (10)$$

Compared to the omitted group of μ change and Markov RGP, only the volatility change has higher RMSE. But all the interactions, except volatility with threshold, have positive coefficients.

8.2 Stylized facts about models

Following the same logic, we can analyze the fixed effects of the models. To capture their sensitivity to mis-specification, we can add interactions between it and the DGP. As all the parameters change in all models, the most important interaction is between the model and the RGP.

Then, the first idea might be to add an indicator of correct specification, but this loses information about the type of mis-specification. Thus, I will also use the full interaction between model and RGP. These are the regressions below.

$$rmse_{p,i,m} = \beta_0 + \beta_1 \text{rgp}_{p,i,m} + \beta_2 \text{sgp}_{p,i} + \beta_3 \text{rgp}_{p,i} \cdot \text{sgp}_{p,i} + \beta_4 \text{mod}_{p,i,m} + \beta_5 \mathbb{1}(\text{mod}_{p,i,m} = \text{rgp}_{p,i}) + \varepsilon_{p,i,m} \quad (11)$$

$$rmse_{p,i,m} = \beta_0 + \beta_1 \text{rgp}_{p,i,m} + \beta_2 \text{sgp}_{p,i} + \beta_3 \text{rgp}_{p,i} \cdot \text{sgp}_{p,i} + \beta_4 \text{mod}_{p,i,m} + \beta_5 \text{mod}_{p,i,m} \cdot \text{rgp}_{p,i} + \varepsilon_{p,i,m} \quad (12)$$

One of the most significant results was a particularly bad interaction between a smooth transition model estimating a structural break RGP.

8.3 Coefficients and performance

The regression above can be expanded to include the dispersion of the estimated coefficients across regimes. This is a similar exercise as the next one, as the coefficients can also be seen as regime-conditional metrics themselves.

8.4 Regimes characteristics and performance

As motivated in the introduction, the characteristics of the regimes, especially the differences between them, could be related to model performance. To analyze this, I propose adding the metrics values as an additional regressor, and an interaction between it and the model. As each metric is specific to each SGP, the regressions will be run separately for each regime nature.

$$rmse_{p,i,m} = \beta_0 + \beta_1 rgp_{p,i,m} + \beta_2 sgp_{p,i} + \beta_3 rgp_{p,i} \cdot sgp_{p,i} + \beta_4 mod_{p,i,m} + \beta_5 mod_{p,i,m} \cdot rgp_{p,i} + \beta_6 c_{p,i,m} + \varepsilon_{p,i,m} \quad (13)$$

In general, the true vs. estimated difference of the conditional metric's dispersion has a positive relation with RMSE.

8.5 Other exercises

The exploratory analysis obtained some 'first step' results, i.e., results on whether the metrics indeed characterize the regimes, and if the models are able to capture that. These could be formalized via regressions too.

The regime variable can be further studied. Its identification performance can be treated as a dependent variable, or also as a control.

An important hyperparameter of the models is the number of regimes $\hat{S}$. To study the sensitivity to mis-specification of this parameter, I can create an indicator variable for $\hat{S} < S$, and check for interactions between it and regime characteristics. The idea is that if the dispersion across regimes is low, mis-specifying the number of regimes should not be as harmful.

If there is time, test if the practical recommendations help in a real-world example, and if the patterns found in the simulations are observed in real data.

9 Conclusion

Here I start by summarizing the motivation and methodology.

Then, I focus on the main results. First, with the more descriptive findings about properties of the models, then, the practical recommendations of metrics an econometrician should look at when choosing a model $\hat{a}$].

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AI disclaimer: this work was generated generally without the help of large language models, except sparringly as a research tool and code autocompletions during the implementation phase.

É necessário colocar algo assim? Talvez como nota de rodapé na primeira página?

A Considered DGPs, models, and metrics

A.1 RGPs and models

A.1.1 Structural Break (SB)

Hypothesis: Regime changes at specific time points $\tau \in (1 : T)^{S-1}$. Formally:

$$\begin{aligned} r_t^s(\cdot ; \tau) &= \mathbb{1}(\tau'_{s-1} < t \leq \tau'_s), \quad \forall s \in \{1, \dots, S\} \\ \tau &\in \mathbb{N}^{S-1}, \quad \tau_s > \tau_{s-1} \quad \forall s, \quad \tau' = (0, \tau, T) \end{aligned} \quad (\text{RGP-SB})$$

Empirical model: Given τ , the model estimates μ and ρ_1 via OLS in each regime. τ is chosen by minimizing the sum of squared residuals over a grid search of breakpoints.

Similarly defined as in (Bai; Perron, 1998). Review of other options in (Casini; Perron, 2018).

A.1.2 Self-Exciting Threshold (SET)

Hypothesis: Regime changes when the series, possibly at a lag $d \in \mathbb{N}^*$, crosses specific threshold values $\tau \in \mathbb{R}^{S-1}$. Transformations of the variable can be considered⁷. Formally:

$$\begin{aligned} r_t^s(\cdot ; (\tau, d, g)) &= \mathbb{1}(\tau'_{s-1} < g(y)_{t-d} \leq \tau'_s), \quad \forall s \in \{1, \dots, S\} \\ \tau &\in \mathbb{R}^{S-1}, \quad \tau_s > \tau_{s-1} \quad \forall s, \quad \tau' = (-\infty, \tau, \infty), \quad d \in \mathbb{N}^* \end{aligned} \quad (\text{RGP-SET})$$

Empirical model: Given τ and d , the model estimates μ and ρ_1 via OLS in each regime. τ and d are chosen by minimizing the sum of squared residuals over a grid search of breakpoints and lags. One can also leave d fixed.

Similarly defined as in (Tong; Lim, 1980). Review of other options in (Chen; Liu; So, 2011).

A.1.3 Smooth Transition (ST)

Hypothesis: Regime changes smoothly, with a continuous function g , often a CDF, based on the difference between the series and the threshold $\tau \in \mathbb{R}$, possibly at a lag $d \in \mathbb{N}^*$. (Medeiros, 2000) has shown that a generalization to S regimes is a neural network, but currently, I only consider $S = 2$. Formally:

$$\begin{aligned} r_t^1(\cdot ; (\tau, d, g)) &= g(y_{t-d} - \tau), \quad r_t^2(\cdot ; (\tau, d, g)) = 1 - r_t^1(\cdot ; (\tau, d, g)) \\ \tau &\in \mathbb{R}, \quad d \in \mathbb{N}^* \end{aligned} \quad (\text{RGP-ST})$$

Often, the function g depends on a smoothness parameter γ , i.e., when $\gamma \rightarrow \infty$, $g \rightarrow \mathbb{1}$. This parameter can be jointly estimated with the others.

Empirical model: Estimated via non-linear squares of the residuals, over μ , ρ_1 (for each regime), τ , and γ . Uses some numerical optimization, which depends on starting values and does not guarantee a global optimum.

Similarly defined as in (Teräsvirta, 1994). Review of other options in (Dijk; Teräsvirta; Franses, 2002)

⁷For example, $g(x) = |x|$ or $g(x) = \Delta x$.

A.1.4 Markov-Switching (MS)

Hypothesis: Regime changes stochastically, following a Markov process with transition matrix $\Gamma \in [0, 1]^{S \times S}$. The probability of being in regime s at time t depends only on the regime at time $t - 1$, often with Γ implying some persistence. Formally:

$$\begin{aligned} r_t^s(\cdot; \Gamma) &\sim P(r_t^s = 1 | r_{t-1}) =: \Gamma_{s, r_{t-1}} \\ \Gamma &\in [0, 1]^{S \times S}, \quad \sum_{i=1}^S \Gamma_{s, i} = 1 \quad \forall s \end{aligned} \quad (\text{RGP-MS})$$

Empirical model: There are multiple algorithms, including maximum likelihood estimation, expectation maximization, and Markov chain Monte Carlo methods. The EM algorithm uses Kalman to find smoothed probabilities of r , then the conditional probabilities given the current guess of parameters, then the guess of parameters is updated via maximizing the likelihood given the probabilities. These two steps are iterated until convergence.

Similarly defined as in (Hamilton, 1989). Review of other options in (Song; Woźniak, 2021).

A.1.5 Unsupervised clustering (UC)

Hypothesis: no hypothesis on the RGP.

Model: Unsupervised clustering techniques, such as K-Means, can be used to estimate the regimes based on y_t , its lags, and rolling moments. Given the regimes, μ and ρ_1 are estimated via OLS. This hybrid approach yields non-standard asymptotic properties. Other clustering techniques could be used, but I focus on the general K-Means clustering problem is as below:

$$\begin{aligned} \hat{r}_t^s(\cdot; (\text{norm}, \text{centroid})) &= \mathbb{1}(y_t \in R_s) \\ R &= \arg \min_{R'} \sum_{s=1}^{\hat{S}} \sum_{y_t \in R'_s} \text{norm}(y_t - \text{centroid}(R'_s)) \end{aligned} \quad (\text{RGP-UC})$$

Similarly defined as in (Akioyamen; Tang; Hussien, 2020). More clustering techniques in (Paparri-zos; Yang; Li, 2024).

A.1.6 Random forests (RF)

Hypothesis: there is no RS, the non-linearity is captured by the tree and ensemble structure of the RF. A review of the time series RF literature is in (Hu; Szymczak, 2022).

Model: a RF is estimated based on y_t , its lags, and rolling moments.

A.2 Metrics

The estimated conditional mean and standard deviation can be calculated as, respectively:

$$\hat{\mu}(y, r | s) := \sum_{t=1}^T r_t^s \cdot y_t \quad (\text{Metric-}\hat{\mu})$$

$$\hat{\sigma}(y, r | s) := \sqrt{\sum_{t=1}^T r_t^s \cdot (y_t - \hat{\mu}(y, r | s))^2} \quad (\text{Metric-}\hat{\sigma})$$

As noted, in the case of binary r_t , only the observations of regime s have non-zero weights, and the formulas are respectively equivalent to:

$$\frac{1}{|R_s|} \sum_{y_t \in R_s} y_t$$

$$\sqrt{\frac{1}{|R_s|} \sum_{y_t \in R_s} (y_t - \hat{\mu}(y|s))^2}$$

For a regime-conditional moments of (y_t, y_{t-j}) , we must define the notion of ‘being in the same regime’. Consider $r_t^s \cdot r_{t-j}^s$, which has a correct ‘truth table’ for binary regimes, but also has an interpretation for continuous ones: closer to 1 the higher the weight of both y_t and y_{t-j} being in regime s . But, this ignores that fact that y_t and y_{t-j} can be in the same regime, but in different regime instances. To account for that, the correct weighting should consider the whole window of $y_{t-j}, \dots, y_t$:

$$\hat{\rho}_j(y, r|s) = \frac{\sum_{t=1+j}^T \left(\prod_{k=1}^j r_k^s \right) \cdot (y_t - \hat{\mu}(y, r|s)) \cdot (y_{t-j} - \hat{\mu}(y, r|s))}{\sum_{t=1+j}^T \left(\prod_{k=1}^j r_k^s \right) \cdot (y_{t-j} - \hat{\mu}(y, r|s))^2} \quad (\text{Metric-}\hat{\rho}_j)$$

For binary regimes, this is equivalent to calculating the unweighted autocorrelation of every concurrent window of regime s .

Recall that a RC metric returns a sequence with entries for each regime, so also consider the following notation:

$$\mu(y_t|S) := (\mu(y_t|S))_{s \in 1:S}$$

A.2.1 True moments of the considered DGPs

Given the weakly stationary within regimes assumption, the regime-conditional moments are independent of the RGP, and are the simple $AR(1)$ moments:

$$\mu(y_t|s) := E[y_t|y_t \in R_s] = \text{frac} \mu^s 1 - \rho_1^s$$

$$\sigma(y_t|s) := \text{Var}[y_t|y_t \in R_s] = \sqrt{\frac{(\sigma^s)^2}{1 - (\rho_1^s)^2}}$$

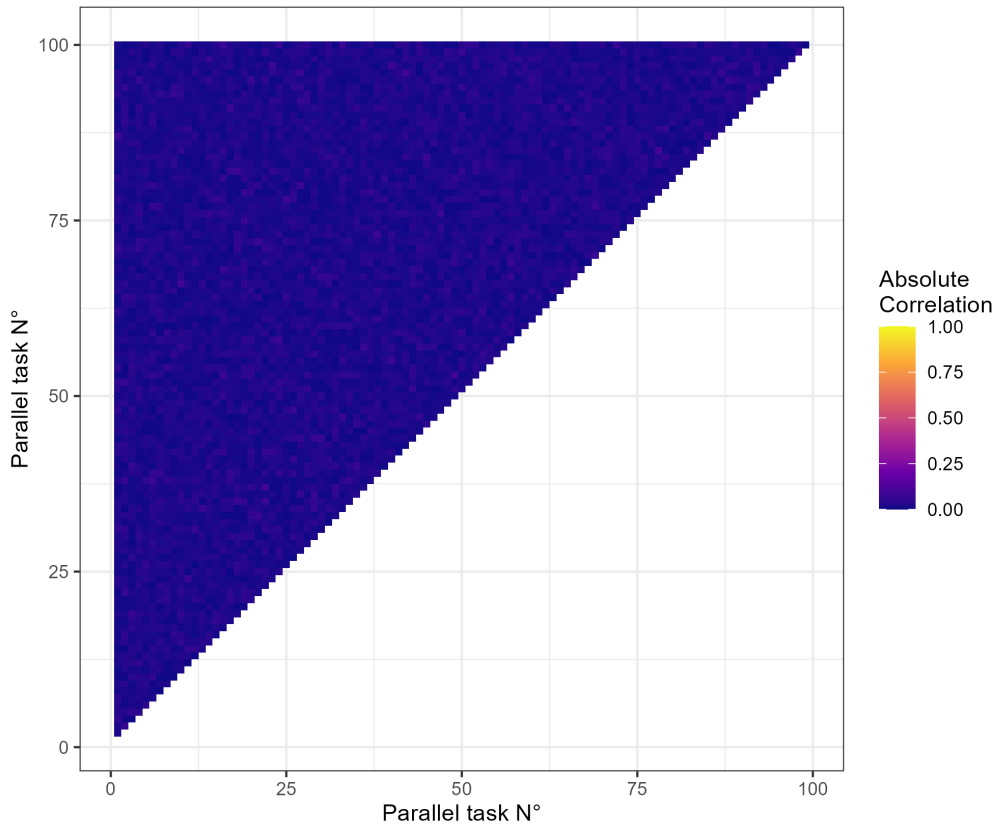
$$\rho_j(y_t|s) := \text{Corr}[y_t, y_{t-1}|y_t \in R_s] = (\rho_1^s)^j, \quad j \in \mathbb{N}^*$$

B Diagnostics

B.1 Random errors

The Figure 9 shows the correlation of the errors across the parallelization structure. A simple visual check shows no evident patterns and an overall low correlation, as expected.

Figure 9: Errors - Correlation across parallelization structure



The Figure 10 shows the distribution of a size 3000 sample of the errors, via the usual histogram and QQ-plot. The distribution is very close to normal, as expected.

B.2 Series generation and model estimation

Table TODO is similar as table TODO, but shows the estimated moments of the model that matched its line's RGP assumption. The values are the average across simulations of the moments calculated with the estimated parameters. Note that the moments don't need to be the exactly the same, since all the models allow for all the parameters to change, a different assumption than of the regime natures.

Figure 10: Errors - Distribution

